molesense

Gel Cathode Design for Electrically Controlled Drug Transport

Parissa Alavi

Transport screening and pillar geometry under fabrication limits

Two design questions

1. Faster, electrically controlled transport

How can distance and gel properties enable rapid transfer
with voltage ON and limited passive redistribution with voltage OFF?

2. Cathode design under fabrication limits

Can shorter, denser pin arrays retain the benefit of the
300 µm design, and how do shape and packing affect the trade-offs?

1

Which gel properties and distance allow rapid transport below 1.23 V?

Goal: increase transfer in 10 minutes at fixed current.

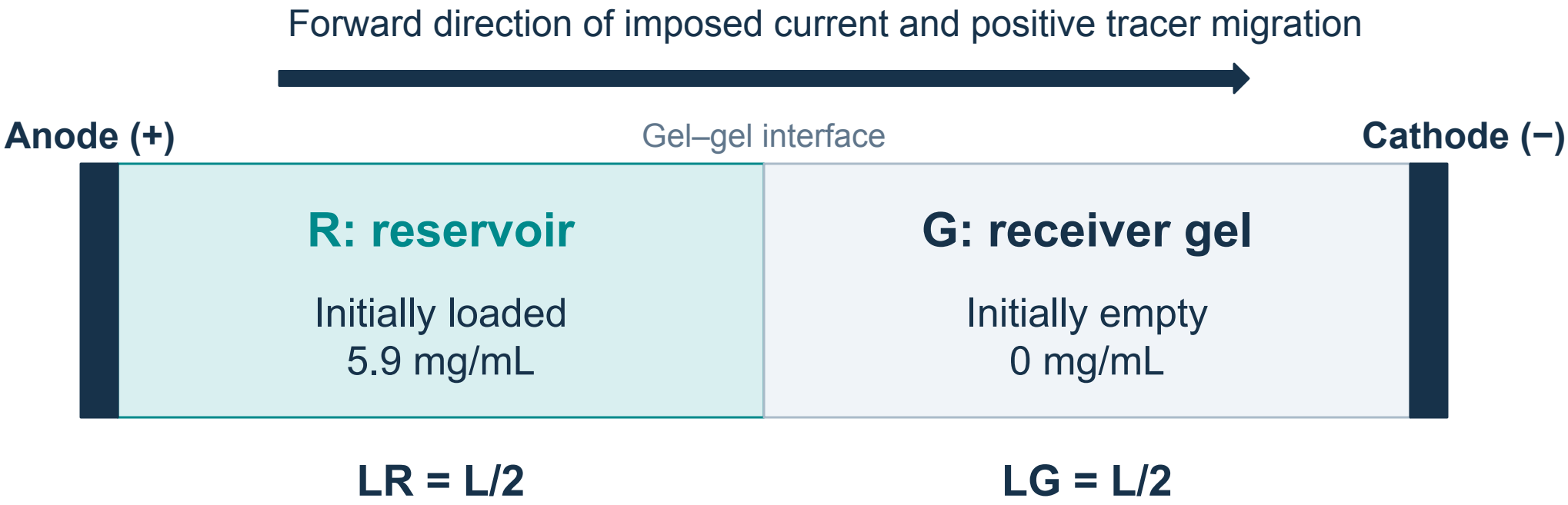
Design variables: diffusivity, conductivity and total path length.

1.23 V is a screening criterion applied to calculated gel voltage.

Parissa Alavi

A fixed-current model links geometry to transport

R denotes the reservoir; G denotes the receiver gel throughout the model.



Total path: L = LR + LG. Cross-section: A = 20 mm² (20 × 1 mm face).

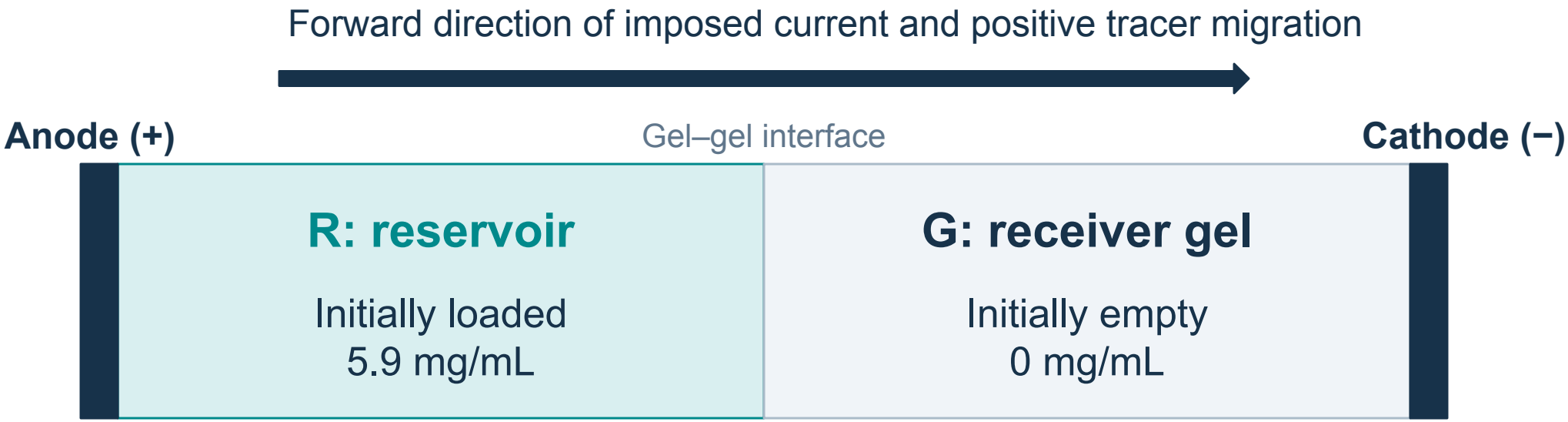
Imposed: 0.1 mA for 10 min

Outputs: receiver fraction and gel voltage

Schematic not to scale. 1D diffusion + electromigration. Gel voltage excludes electrode/contact losses.

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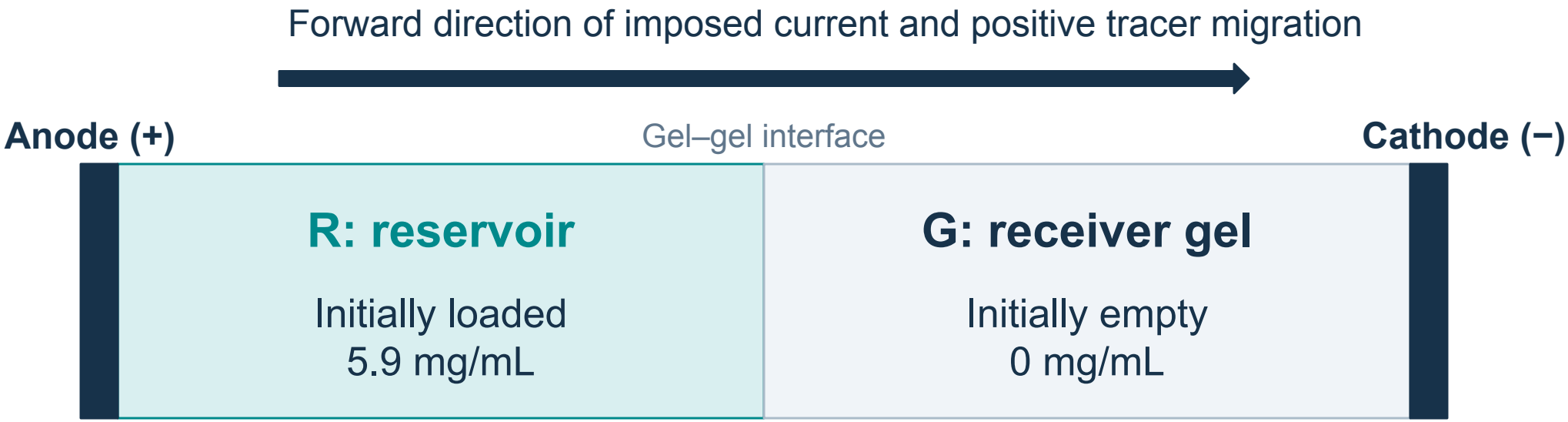


$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x}$$

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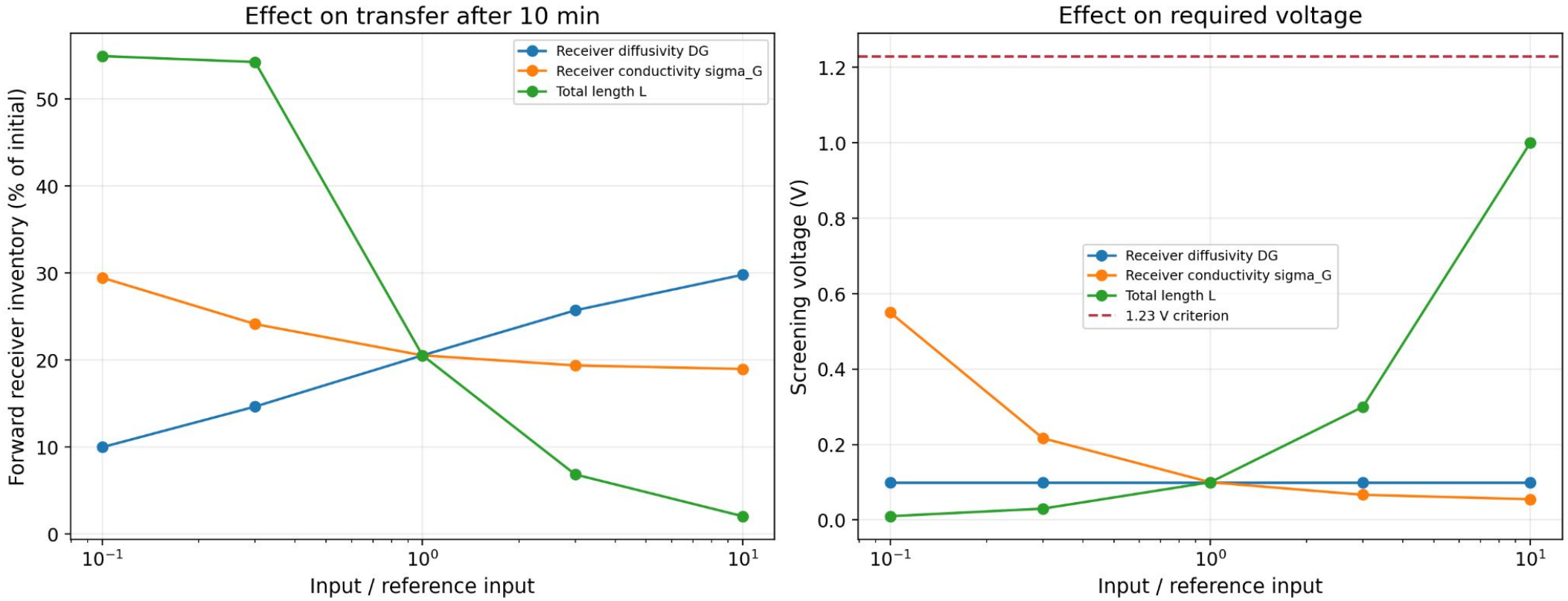
Diffusion + electromigration

$$J = -D(x)\frac{\partial c}{\partial x} + \frac{zFD(x)}{RT}\,c(x,t)\,E(x)$$

Schematic not to scale. 1D diffusion + electromigration. Gel voltage excludes electrode/contact losses.

Length and receiver properties affect transport differently

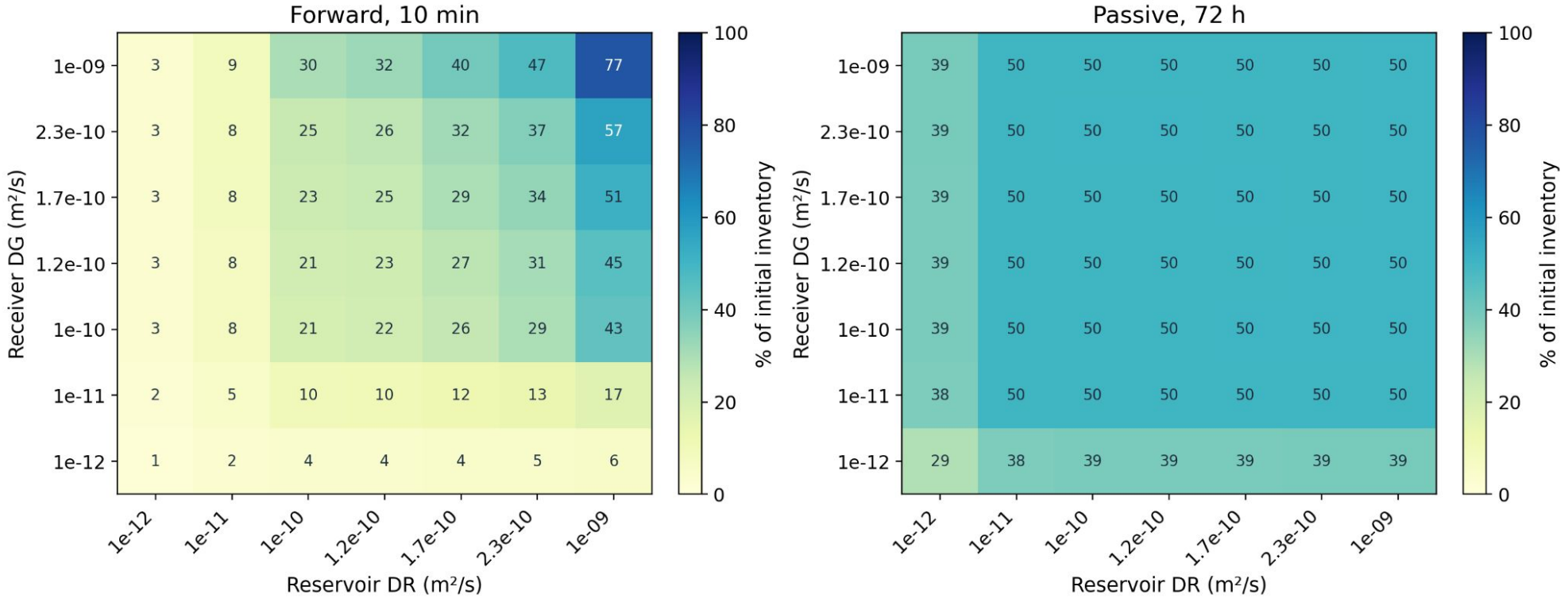
One input varies at a time; reservoir properties and imposed current stay fixed.



Shortening the path raises transfer and lowers gel voltage. Higher receiver conductivity lowers voltage, but also reduces the local electric field.

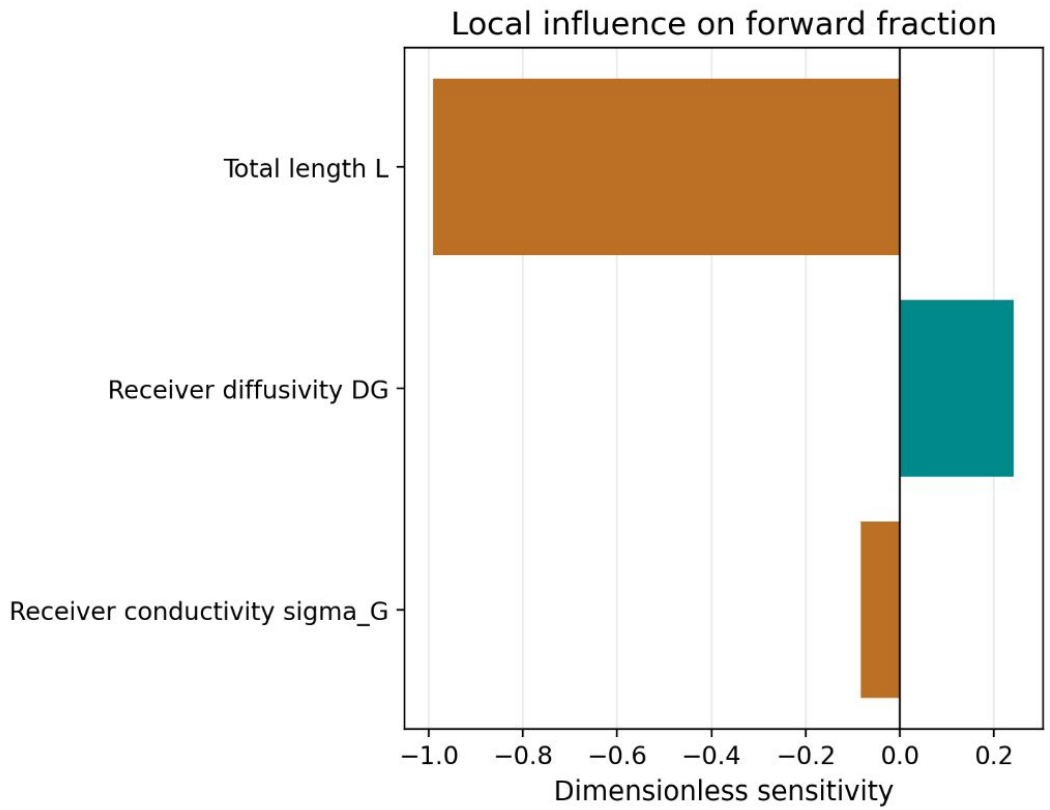
How diffusivity in the reservoir and receiver affects transport

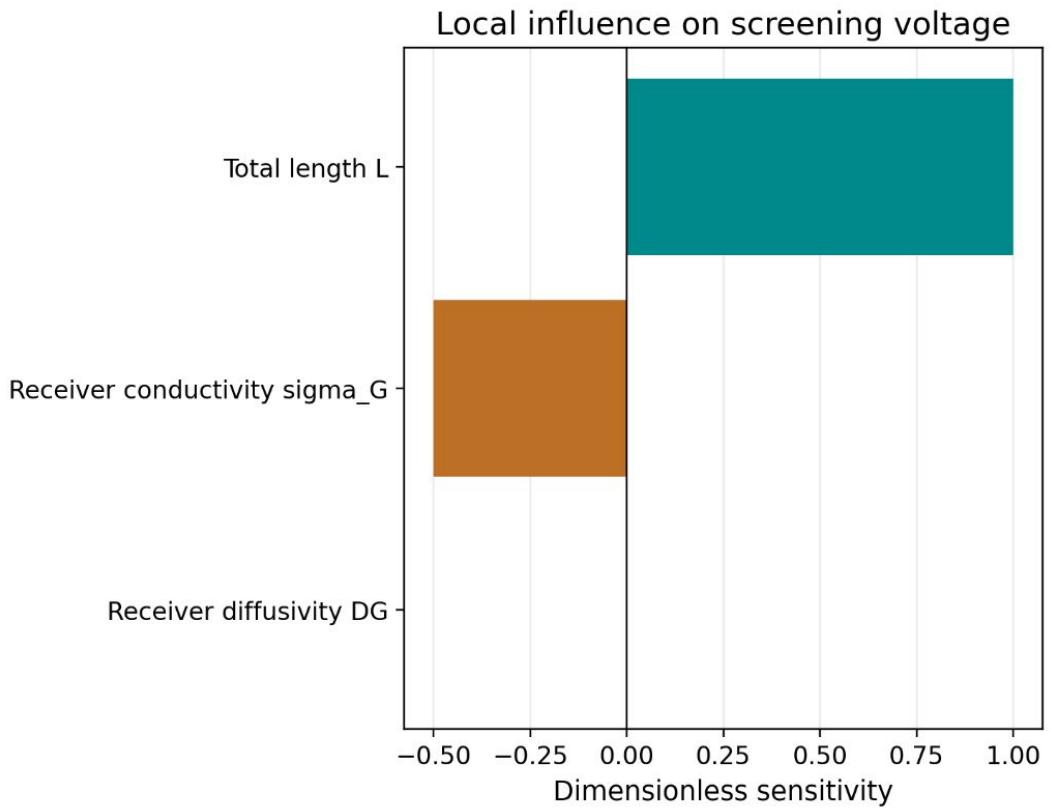
The numbers show the percentage of the initial drug inventory reaching the receiver.
diffusivity also determines electrical mobility, so this increase includes both diffusion and electromigration



Length has the strongest local influence near the reference

Sensitivity compares equal relative changes: ±10% around the same reference case.

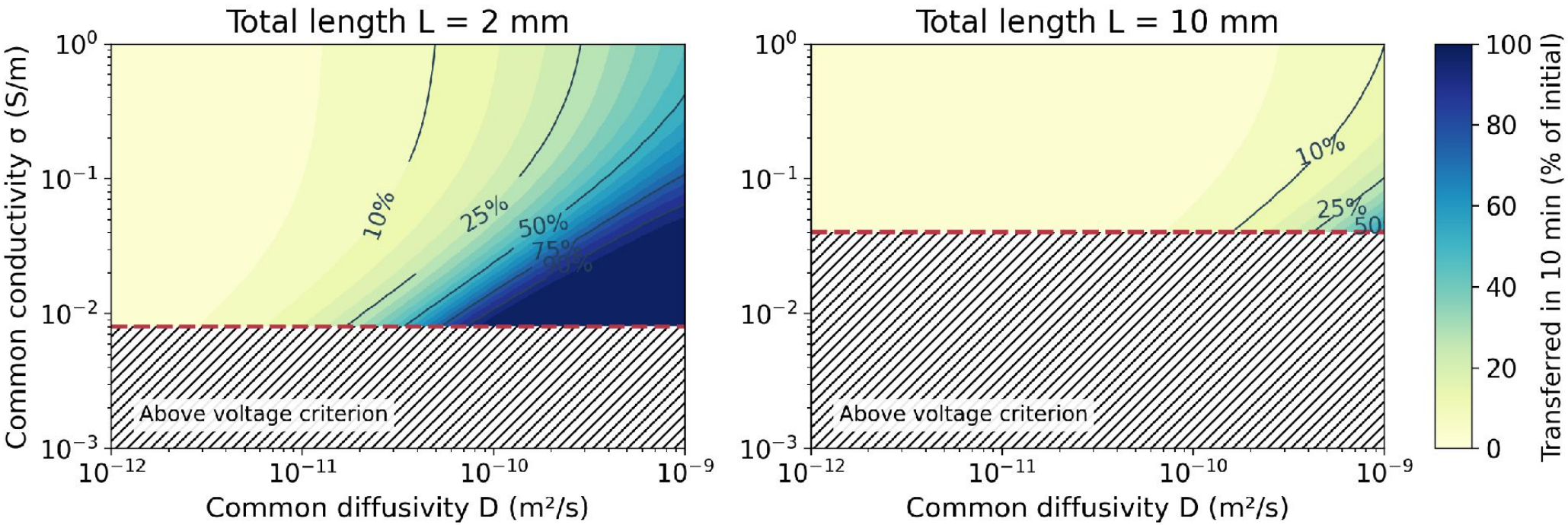




Near this reference, length dominates transfer and voltage; receiver diffusivity affects transfer, while conductivity mainly reduces voltage.

The contour map identifies voltage-feasible combinations

Identical-gel slice: DR = DG = D and σR = σG = σ; A = 20 mm², I = 0.1 mA



Colours and labels: percentage transferred in 10 minutes.

Red boundary: 1.23 V. Hatching: voltage criterion exceeded.

Next: calibrate the model, then test selected candidates

1. Measure conductivity and device voltage

Use the same current and geometry to quantify bulk and electrode/contact contributions.

2. Define the required transferred amount and time

Choose the delivery target before calling a combination acceptable.

3. Test candidates selected from the contour map

Compare measured transferred mass and device voltage with the model predictions.

2

Height limits motivate shorter, denser arrays

Can a shorter, denser array retain the benefit of the 300 µm design?

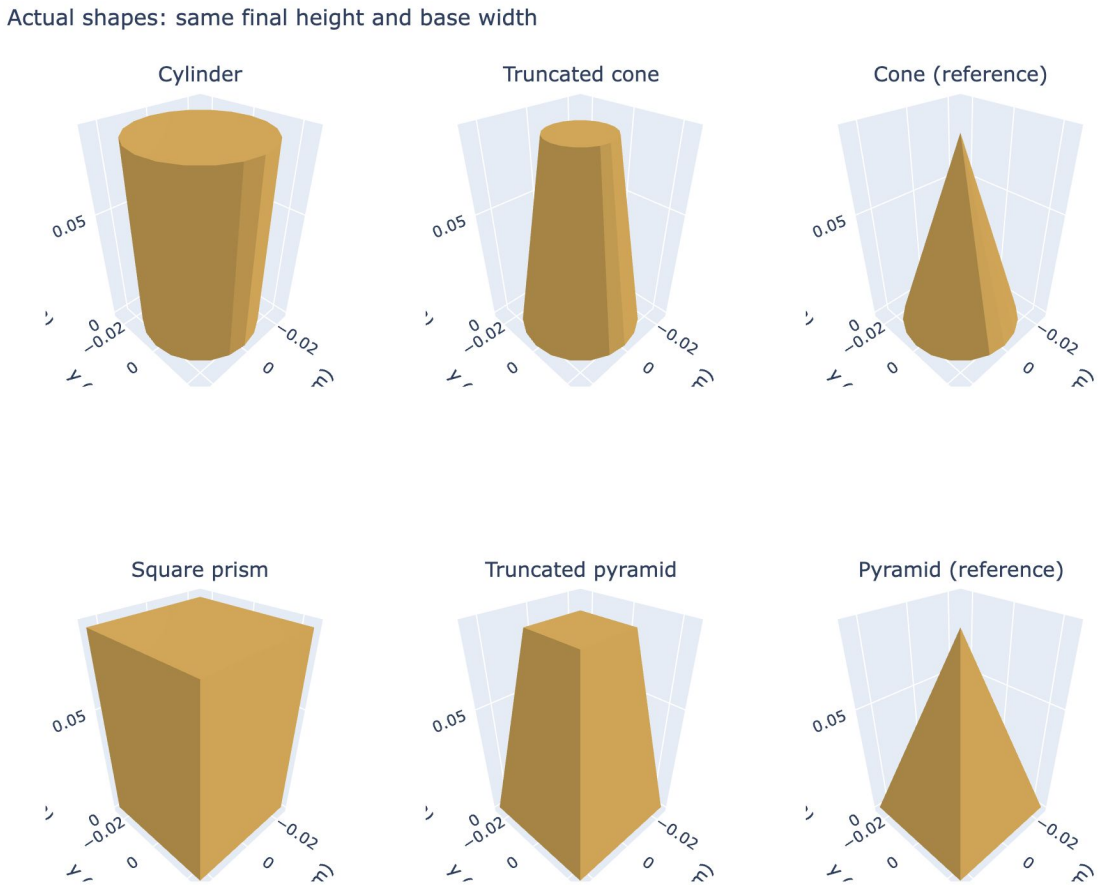
The 300 µm design provides the reference. Fabrication motivates exploring heights below 100 µm.

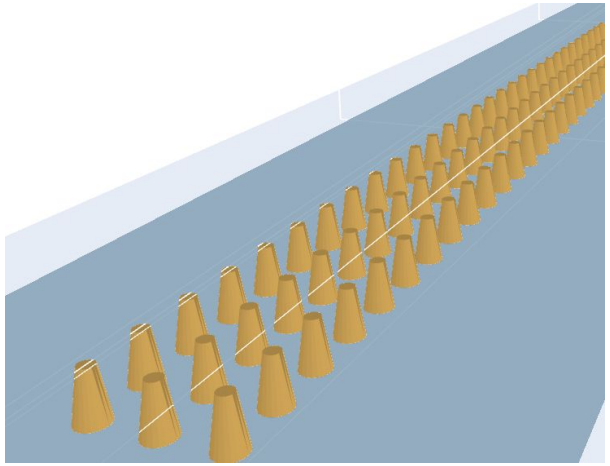
This study screens surface area. COMSOL will evaluate voltage and resistance.

Ideal geometry • 20 × 1 mm assumed electrode • results require electrical validation

Six shapes isolate the effects of profile and cross-section

The six shapes and array view define the geometry and packing comparison.

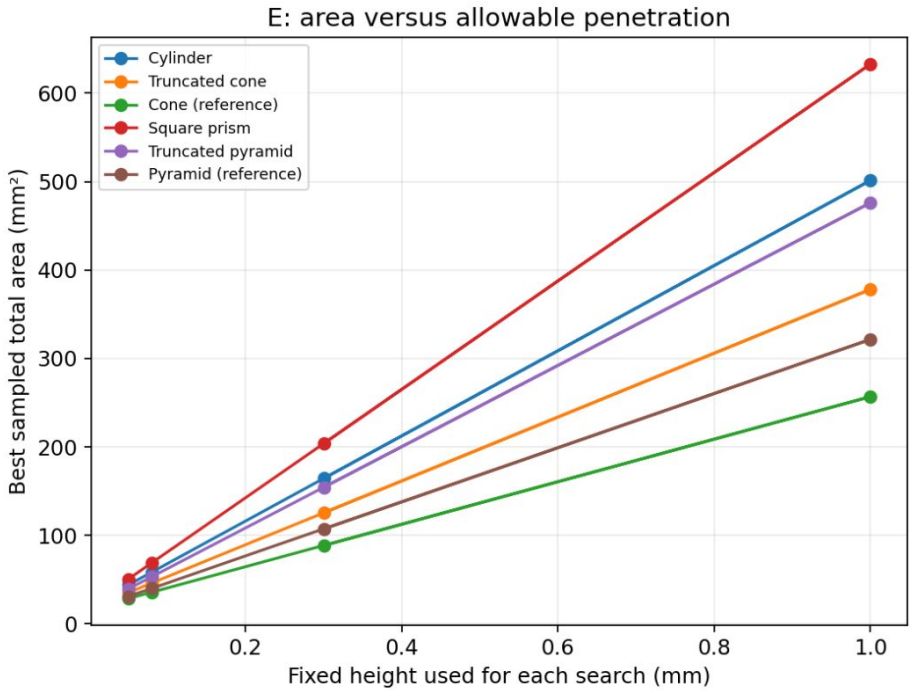


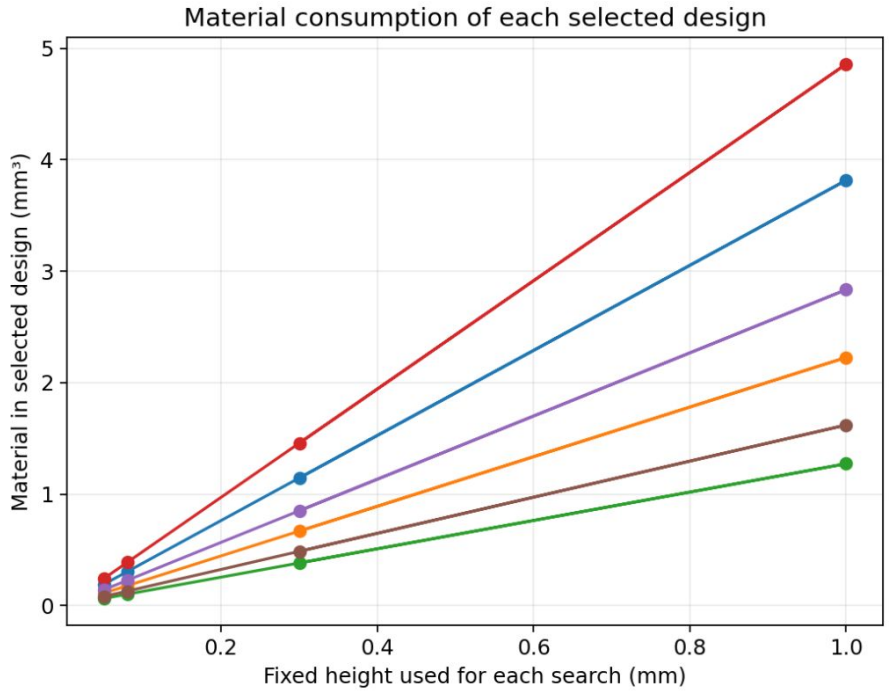


Shape comparison

Same h, width and N
Material volume varies

More allowable height enables more exposed area





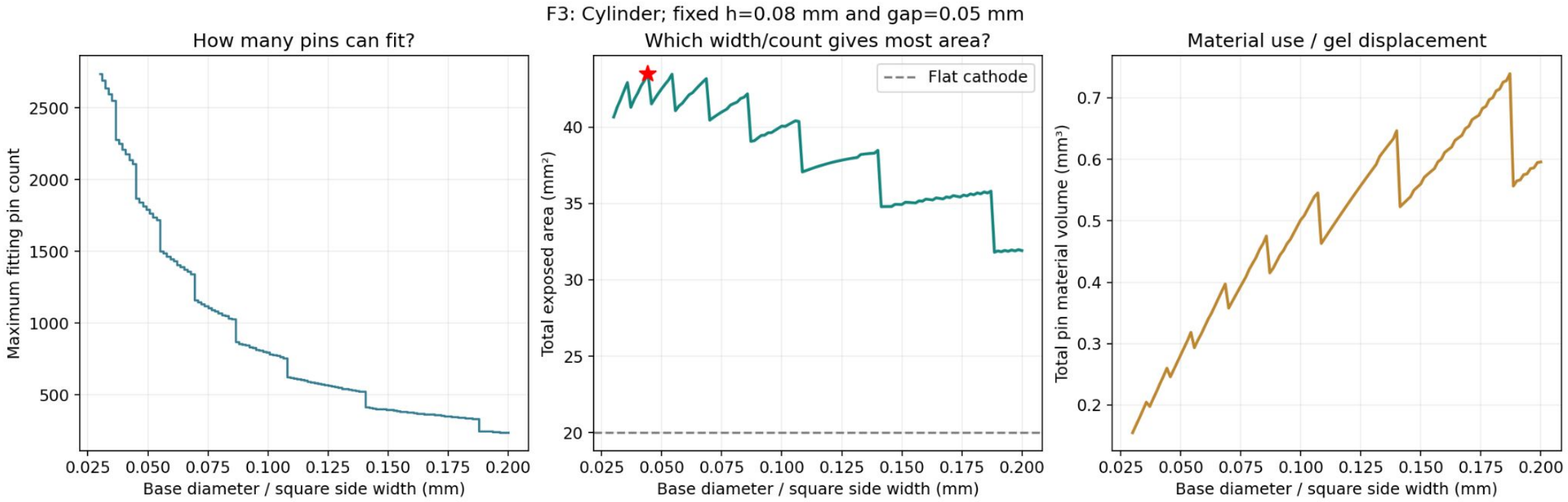
At each height the plotted result is the best sampled geometry for each shape.

Cylinder: 44.1 mm² at 50 µm and 58.5 mm² at 80 µm.

The plotted tall designs also use assumed bounds; area is not resistance.

Ideal geometry • 20 × 1 mm assumed electrode • results require electrical validation

Wider pillars leave room for fewer pillars

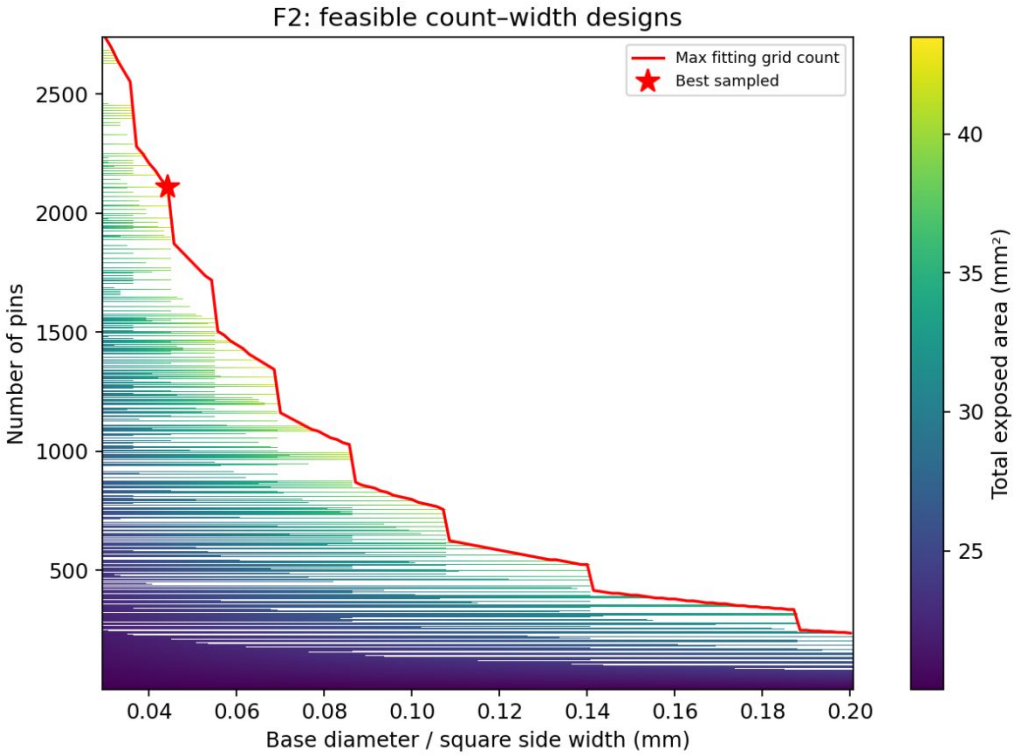


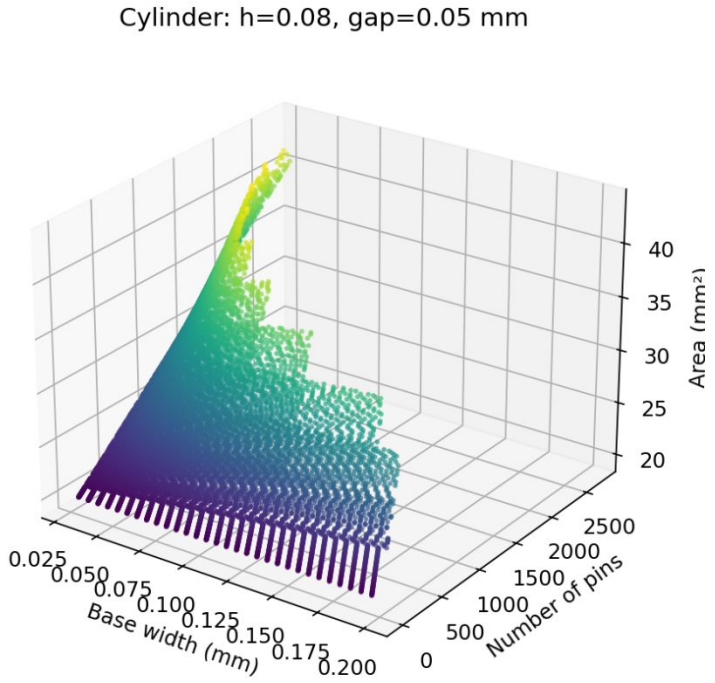
At 80 µm height and 50 µm gap, larger pillars reduce how many fit. Steps occur when a row or column no longer fits.

The peak is an area maximum under these assumptions, not an electrical optimum.

Ideal geometry • 20 × 1 mm assumed electrode • results require electrical validation

Packing limits the feasible width–count combinations





Cylinders: 80 µm height and 50 µm gap

The red line shows the maximum fitting rectangular grid. The star marks the best sampled area.

The full-packing boundary constraints how diameter and count can combine.

Dense short arrays recover part of the reference area

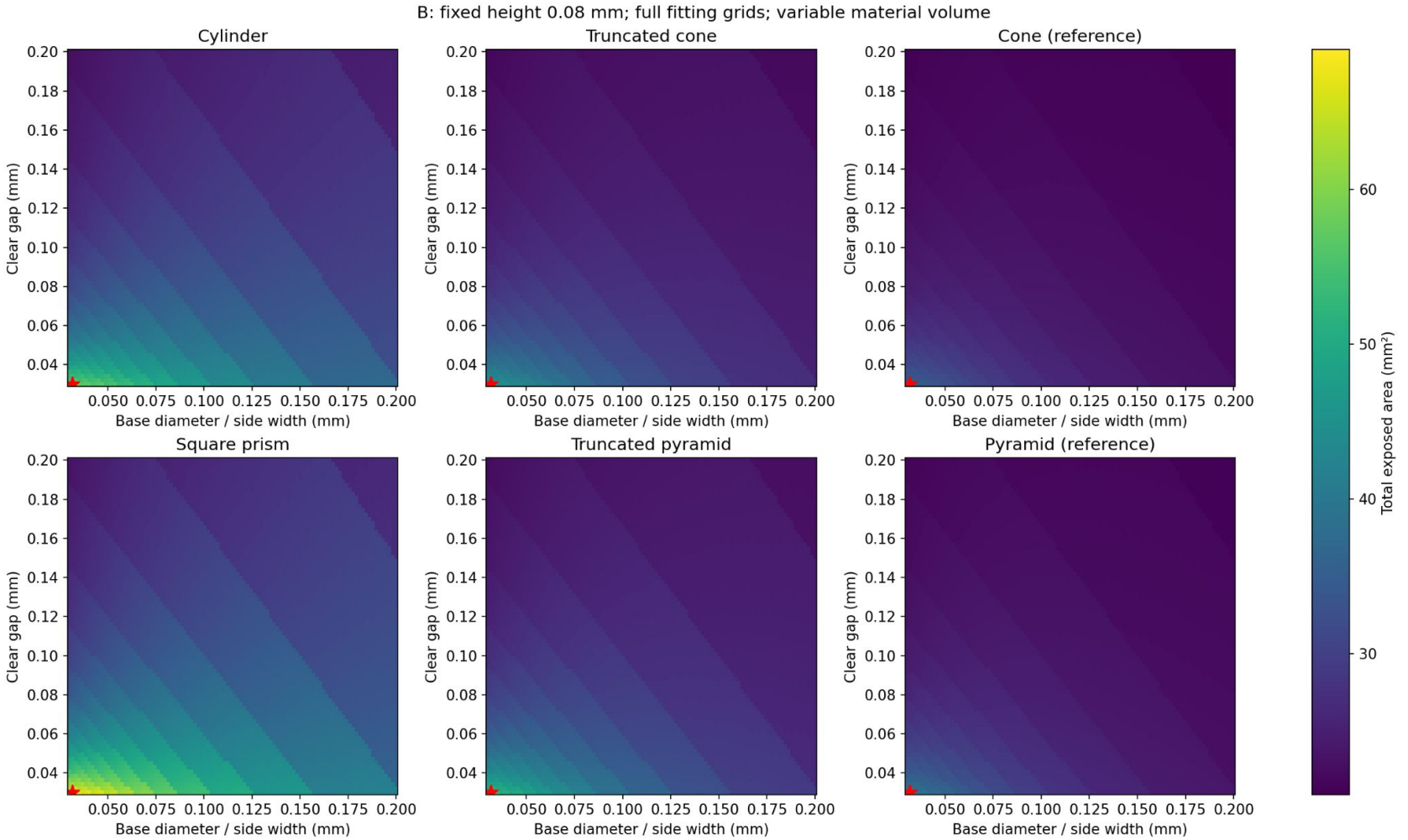
20 × 1 mm face, 50 µm margins, complete rectangular arrays

Cylinder array	h (µm)	d (µm)	Gap (µm)	Count	Area (mm²)
Tall reference	300	50.0	50	1,791	104.4
Short, same layout	80	50.0	50	1,791	42.5
Short, denser array	80	31.7	30	4,830	58.5

58.5 / 104.4 ≈ 56% of total exposed area

Thank you for your attention

Smaller gaps increase the area that fits



At 80 µm height, these maps show how width and gap change the maximum area that fits.

Cylinder maximum: d ≈ 31.7 µm, gap 30 µm, 4,830 pillars, 58.5 mm².

30 µm is the search boundary, not a validated fabrication limit.

Ideal geometry • 20 × 1 mm assumed electrode • results require electrical validation

NEXT STEP

Proposed COMSOL comparison

Geometry	Height	Diameter	Pitch
Flat reference	0	—	—
Existing cylinder	300 µm	50 µm	100 µm
Short cylinder	80 µm	50 µm	100 µm
Denser candidate	80 µm	≈32 µm	≈62 µm

Compare voltage at the same current density and physical electrode separation. Check the minimum gap with fabrication.

Ideal geometry • 20 × 1 mm assumed electrode • results require electrical validation